

Yuncong Yu



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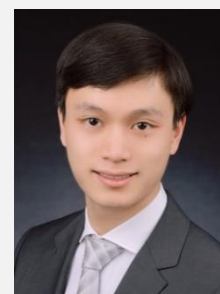
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Max-Planck-Str. 12, 76344 Eggenstein-Leopoldshafen, Germany



Work Experience

01.2020 – Present



IAV GmbH Ingenieurgesellschaft Auto und Verkehr, Gifhorn

Data Scientist | Department of Data Science & AI Projects

- Development of agent-based AI solutions to automate engineering workflows and orchestrate internal tool ecosystems.
- Analysis of battery aging effect to improve prediction accuracy and uncertainty quantification with Bayesian inference
- Development of a digital twin for modeling non-exhaust emissions based on Physics-Informed Neural Networks (PINNs).
- Architecture design and algorithm development for an analytics platform with a focus on anomaly detection and pattern recognition.
- Application of Large Language Models (LLMs) for automated requirements analysis and AI-assisted test case generation.

03.2019 – 12.2019



TLK-Thermo GmbH, Brunswick/Ingolstadt

Development Engineer

- Co-Simulation, optimization, and control of fuel cell systems with MATLAB/Simulink and Modelica/Dymola.
- Data analysis and process automation with Python.

07.2018 – 12.2018



AUDI AG, Ingolstadt

Master Thesis | Department of Thermal Management for Fully Electric Vehicles

- Data-driven analysis of multivariate time series with Python and TensorFlow.
- Validation and integration of thermal management functions with MATLAB/Simulink.

10.2017 – 03.2018

Internship | Department of Engine Flow and Aerodynamics Simulation

- Optimization and validation of an automotive climatization model.
- Co-simulation with MATLAB/Simulink and CFD simulation with OpenFOAM.

Education

12.2020 – 01.2025



Utrecht University, Utrecht, the Netherlands

Computer Science (Ph.D.)

- Research focus: time series analysis and visual analytics.
- Doctoral Thesis: Multivariate Time Series Pattern Search.

09.2015 – 12.2018



Karlsruhe Institute of Technology (KIT), Karlsruhe

Mechanical Engineering (M.Sc.)

- Focus areas: mechatronics and Simulation.
- Final grade: 1.4.
- Master's thesis: Segmentation, Comparison, and Interpretation of Multivariate Time Series (grade: 1.0).

09.2011 – 06.2015



Sichuan University (SCU), Chengdu, China

Mechanical Engineering (B.Eng.)

- Final grade: 92/100; ranked 1st out of 358 students.
- Bachelor's thesis: Design and Finite Element Analysis of a Synthetic Human Organ Injection Mold (The first price of excellent bachelor thesis).
- National Scholarship x2, outstanding graduate student of the university.

Practical Experiences

05.2017 – 07.2017



Karlsruhe Institute of Technology (KIT), Karlsruhe

Cognitive Automobile Lab | Institute of Measurement and Control Technology (MRT)

- Development of autonomous driving functions in a six-person team.
- Perception as well as longitudinal and lateral control in ROS.

10.2016 – 03.2017

Scientific Assistant | Institute of Technical Mechanics (ITM)

- Tutor for computer-aided simulation with MATLAB/Simulink
- Topics: numerics, NVH, CFD, FEM, and DoE

06.2013 – 06.2014



SCU, Faculty of Mechanical Engineering, Chengdu, China

Project | National University Competition of Mechanical Innovation Design

- Team leader of 4 students
- The second national price and the first provincial price

Selected Publications

Y. Yu, T. Becker, P. Schichtel, W. Aigner, A. Telea, Michael Behrisch; NOOPS: Nonmyopic Feedback-Driven Pattern Search in Multivariate Time Series. <http://dx.doi.org/10.2139/ssrn.5180352>

Y. Yu, T. Becker, W. Aigner, A. Telea, and M. Behrisch; Time Series Representation Techniques: a Survey. <http://dx.doi.org/10.2139/ssrn.5180323>

Y. Yu, T. Becker, L. M. Trinh, M. Behrisch; SAXRegEx: Multivariate Time Series Pattern Search With Symbolic Representation, Regular Expression, and Query Expansion. In Computers & Graphics 112. DOI: 10.1016/j.cag.2023.03.002. **(Best Paper in EuroVA 2022)**

Y. Yu; D. Kruffy; J. Jiao; T. Becker; M. Behrisch; PSEUDO: Interactive Pattern Search in Multivariate Time Series with Locality-Sensitive Hashing and Relevance Feedback. In IEEE Transactions on Visualization and Computer Graphics 29 (1). DOI: 10.1109/TVCG.2022.3209431. **(Best Paper in VDS 2022)**

Language Skills

German	fluent	(TestDaF: 18/20)
English	fluent	(TOEFL: 100/120, GRE: 322/340 3.5/6, CET-6: 602/720)
Chinese	first language	

Computer Skills

Data Science & AI: PyTorch, TensorFlow, Scikit-Learn, Numpy, LangChain
Programming Languages: Python, MATLAB/Simulink, C/C++, JavaScript/TypeScript
WebDev & Tool Development: React, Django, FastAPI, MySQL, PostgreSQL, D3
DevOps: Git, Docker, Kubernetes, CI/CD
Simulation & Engineering: Dymola/Modelica, ROS, CATIA, AutoCAD

Others

Driving licence in China and Germany, Prototype driving licence